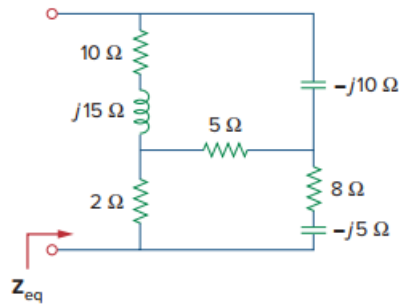


Problem

Find the equivalent impedance of the circuit in Fig. 9.77.

Figure 9.77:

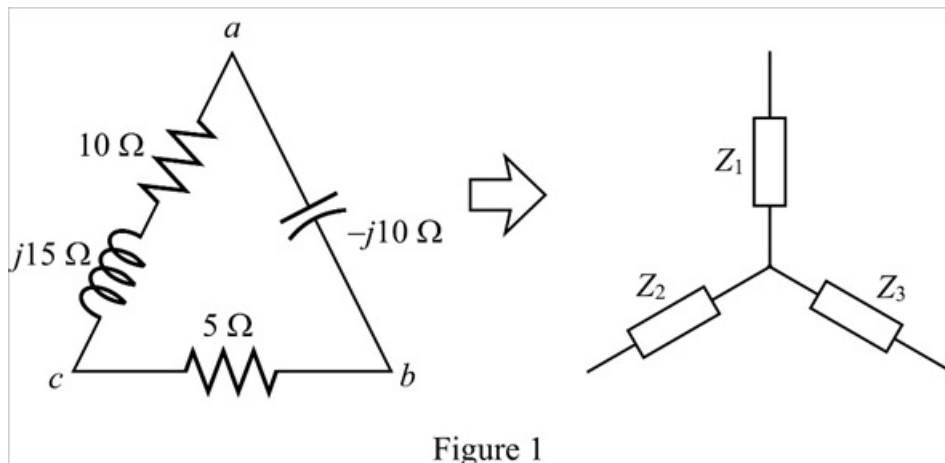


Step-by-step solution

Step 1 of 4

Refer to circuit diagram in Figure 9.77 in the textbook.

In the circuit, $-j10\ \Omega$ branch, $5\ \Omega$ branch and series combination of $10\ \Omega$ and $j15\ \Omega$ branch are connected in delta. Convert this delta connection into equivalent wye connection. The circuit of converting delta to wye is shown in Figure 1.



Step 2 of 4

Calculate the impedance Z_1 .

$$\begin{aligned} Z_1 &= \frac{Z_{ab}Z_{ca}}{Z_{ab} + Z_{bc} + Z_{ca}} \\ &= \frac{(-j10)(10 + j15)}{-j10 + 10 + j15 + 5} \\ &= \frac{150 - j100}{15 + j5} \\ &= 7 - j9 \, \Omega \end{aligned}$$

Calculate the impedance Z_2 .

$$\begin{aligned} Z_2 &= \frac{Z_{bc}Z_{ca}}{Z_{ab} + Z_{bc} + Z_{ca}} \\ &= \frac{(5)(10 + j15)}{-j10 + 10 + j15 + 5} \\ &= \frac{50 + j75}{15 + j5} \\ &= 4.5 + j3.5 \, \Omega \end{aligned}$$

Calculate the impedance Z_3 .

$$\begin{aligned} Z_3 &= \frac{Z_{bc}Z_{ab}}{Z_{ab} + Z_{bc} + Z_{ca}} \\ &= \frac{(5)(-j10)}{-j10 + 10 + j15 + 5} \\ &= \frac{-j50}{15 + j5} \\ &= -1 - j3 \, \Omega \end{aligned}$$

Step 3 of 4

Draw the modified circuit diagram.

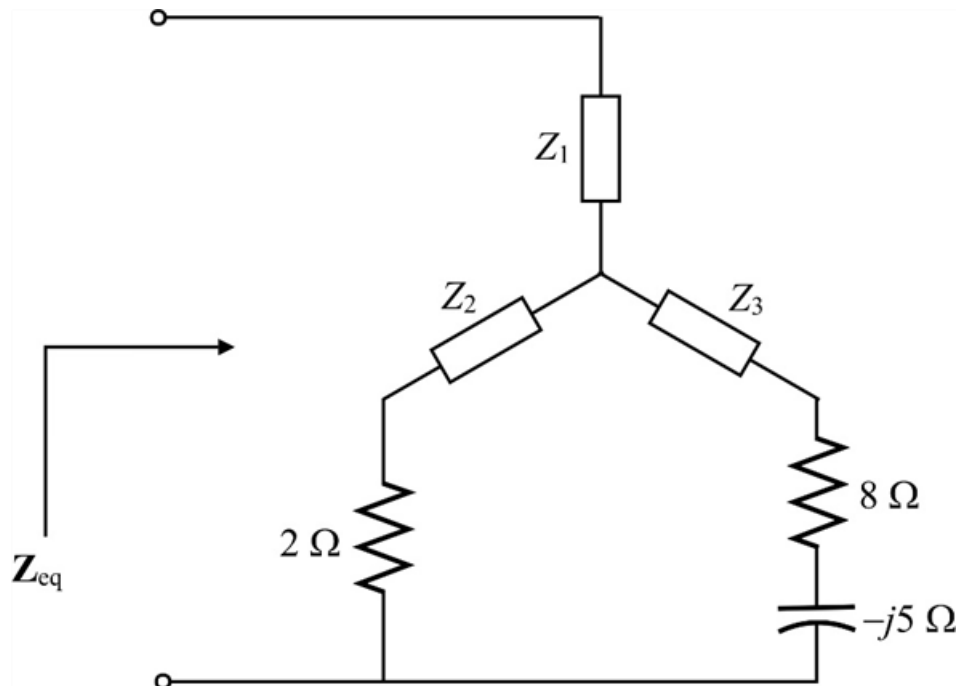


Figure 2

Step 4 of 4

Calculate the equivalent impedance from the circuit shown in Figure 2.

$$\mathbf{Z_{eq}} = \mathbf{Z_1} + [\mathbf{Z_2} + 2] \mathbf{P}[\mathbf{Z_3} + 8 - j5]$$

Substitute $7 - j9 \, \Omega$ for $\mathbf{Z_1}$, $4.5 + j3.5 \, \Omega$ for $\mathbf{Z_2}$ and $-1 - j3 \, \Omega$ for $\mathbf{Z_3}$.

$$\mathbf{Z_{eq}} = 7 - j9 + [4.5 + j3.5 + 2] \mathbf{P}[-1 - j3 + 8 - j5]$$

$$= 7 - j9 + \frac{(6.5 + j3.5)(7 - j8)}{6.5 + j3.5 + 7 - j8}$$

$$= 7 - j9 + 5.511 - j0.2$$

$$= 12.511 - j9.2 \, \Omega$$

$$= 15.53 \angle -36.33^\circ \, \Omega$$

Therefore, the value of equivalent impedance is $\boxed{15.53 \angle -36.33^\circ \, \Omega}$.